Yemen Conflict and Global Trade

GS II

- Yemen lies beside the Bab el-Mandeb Strait, a narrow passage connecting the Red Sea with important international shipping routes. Fighting near this **maritime chokepoint** can endanger commercial vessels and disrupt the movement of goods between regions.
- The conflict becomes more dangerous because the Houthis receive Iranian support while the internationally recognised Yemeni government receives Saudi backing. This **external involvement** connects Yemen's internal struggle with wider regional rivalries and security concerns.
- Attacks on ships and energy infrastructure can force vessels to avoid dangerous waters, increasing **transportation costs** and delaying supplies. Disruptions become especially serious when other important routes, including the Strait of Hormuz, face restrictions.
- Military operations alone have struggled to end the fighting because the opposing groups retain political and military interests. A lasting settlement requires negotiations within Yemen alongside efforts to reduce **regional tensions** and protect maritime security.

Yemen's conflict demonstrates how internal instability, regional rivalries and vulnerable shipping routes can combine to threaten international commerce, making political settlement essential for lasting security.

Neutrinos and Cosmic Exploration

GS III

- Ordinary telescopes study **electromagnetic radiation**, including visible light, but some cosmic environments are too dense for light to escape. Scientists therefore need other messengers capable of carrying information from these otherwise hidden regions of the universe.
- Neutrinos provide such a messenger because they have extremely small masses, no electric charge and interact very **weakly with matter**. They can travel through dense material while preserving information about the extreme environments where they originated.
- The IceCube Observatory uses Antarctic ice to detect rare neutrino interactions. When a neutrino occasionally interacts with matter, it can produce charged particles that generate **faint flashes of light**. Thousands of embedded sensors help scientists reconstruct its direction and energy.
- By analysing these detections, researchers can investigate energetic cosmic events and identify possible sources of high-energy neutrinos. Combining **neutrino observations** with conventional astronomy creates a broader understanding of physical processes occurring across the universe.

Neutrino astronomy allows scientists to investigate cosmic environments beyond the reach of ordinary telescopes, expanding our understanding of energetic events and the physical universe.

RBI and Inflation Control

GS III

- Inflation can rise when essential supplies become expensive or scarce. Higher crude oil prices increase energy costs, while **poor rainfall** can reduce agricultural production, pushing food prices upwards even without stronger consumer demand.
- The RBI raises its policy interest rate to make borrowing more expensive and moderate spending. However, **monetary tightening** cannot directly produce more food or reduce international oil prices, limiting its immediate effectiveness against supply-driven inflation.
- **Inflation expectations** create another challenge. When households and businesses anticipate continuing price increases, they may demand higher wages or raise selling prices. Such behaviour can sustain inflation even after the original supply disruption begins to ease.
- Controlling inflation therefore requires **coordinated action**. The RBI can manage demand and expectations, while the government improves food availability through buffer stocks, trade measures and market intervention. Both must consider the impact on economic growth.

Interest rate increases cannot fully resolve inflation caused by supply shortages. Effective price stability requires monetary policy, government supply management and careful protection of economic growth.

Molecular Chirality and Its Importance

GS III

- Some molecules exist in two mirror-image arrangements that cannot be perfectly superimposed, much like human hands. This property is called **chirality**, and it means molecules with identical chemical compositions can possess different three-dimensional structures.
- Living organisms contain biological molecules with specific shapes. Their **enzymes and receptors** may interact differently with each mirror-image form, allowing one version to participate effectively in a biological process while another behaves differently.
- This difference becomes especially important in pharmaceuticals because molecular shape can influence **therapeutic effects** and safety. Scientists must understand how each form behaves rather than assuming that mirror-image molecules will produce identical biological responses.
- **Asymmetric synthesis** helps chemists produce a preferred molecular form. Certain reactions can also amplify a small initial excess of one form, offering insights into biological homochirality and supporting more precise methods of manufacturing useful chemicals.

Molecular chirality shows why three-dimensional structure matters in biology and medicine, while selective chemical synthesis improves pharmaceutical development and helps investigate the molecular origins of life.